Band-1 Computational Experiment: Erdős–Straus Conjecture Finite Verification

Human + ZoraASI full-dash (executor)

2026-09-11

Band-1 Computational Experiment: Erdős–Straus Finite Verification

Honesty stamp. This note **verifies** that for every integer n with $2 \le n \le N$ there exist positive integers $a \le b \le c$ satisfying

$$\frac{4}{n} = \frac{1}{a} + \frac{1}{b} + \frac{1}{c}$$

for an explicit finite N computed on this box. It does **not** prove the **Erdős–Straus conjecture** (the same statement for **all** integers $n \ge 2$). Finite-range verification $\neq$ conjecture. **Conjecture status: OPEN.** This is **not** a Clay Millennium claim and **not** a Theory-of-Everything claim.

Abstract

We implement modular Egyptian-fraction identities (Mordell-style), composite lifting from prime factors, and a shallow-a divisor search for residual n, and verify Erdős–Straus representations on the full range

$$\mathbf{N = 1000000 = 10^6}.$$

All **999999** integers $n \in [2, N]$ admit a solution (failures: **0**). Wall time **18.27 s**. Literature computational bounds reach 10^{17} (Salez) and a 2025 preprint reports 10^{18}; both far exceed this box probe. **Conjecture status: OPEN.**

Problem statement

Erdős–Straus conjecture (1948). For every integer $n \ge 2$ there exist positive integers a, b, c with $4/n = 1/a + 1/b + 1/c$.

This experiment. Prove computationally that the statement holds for all $n \in [2, N]$ for a box-feasible N (minutes), record sample solutions, and ledger the literature status (OPEN; large verified bounds).

Equivalent Diophantine form: $4abc = n(ab + ac + bc)$.

Literature ledger (not re-proved on box)

Year	Limit / note	Authors	Status
1950	$5 \cdot 10^3$ / $2 \cdot 10^4$ / 10^5	Straus / Shapiro / Obláth	historical
1967	mod-840 identities (6 uncovered classes)	Mordell	theory

Year	Limit / note	Authors	Status
2013	average solution count polylog	Elsholtz & Tao	theory
2014	10^{17}	Salez, arXiv:1406.6307	preprint (widely cited)
2025	10^{18} reported	Mihnea & Bogdan, arXiv:2509.00128	preprint
2026	remains OPEN	e.g. arXiv:2608.24035	open

Primary computational citations: - S. E. Salez, *The Erdős-Straus conjecture: New modular equations and checking up to* $N = 10^{17}$, arXiv:1406.6307 (2014). - S. Mihnea, D. C. Bogdan, *Further verification and empirical evidence for the Erdős-Straus conjecture*, arXiv:2509.00128 (2025). - Wikipedia *Erdős–Straus conjecture* (Computational results) cites $n \le 10^{17}$.

It is enough to search **primes** for counterexamples (composite $n = mp$ lifts from a solution for p). This box verifies **all** $n \in [2, N]$ with explicit (a, b, c).

Modular obstruction patterns (brief)

Mordell (1967) gave polynomial identities covering n in many residue classes (e.g. 2 (mod 3), 3 (mod 4), 2 or 3 (mod 5), $3, 5, 6$ (mod 7), 5 (mod 8)). Combining them leaves at most the six classes

$$n \equiv 1, 121, 169, 289, 361, 529 \pmod{840}$$

as possible identity-gaps. **No complete covering system exists**: an identity for all $n \equiv r \pmod m$ can exist only when r is **not** a quadratic residue mod m, but 1 is always a square, so $r = 1$ is forever uncovered (Mordell nonexistence). Potential counterexample density is nonetheless 0 (Webb and subsequent density results). The greedy Egyptian algorithm already uses ≤ 3 terms except when $n \equiv 1$ or 17 (mod 24); the 17 (mod 24) case sits inside 2 (mod 3) and is covered by an identity.

Method (box)

1. **Identities.** Apply verified constructive identities for even n; $n \equiv 3 \pmod 4$; $n \equiv 2 \pmod 3$; $n \equiv 5 \pmod 8$; $n \equiv 2, 3 \pmod 5$; $n \equiv 3, 5, 6 \pmod 7$.
2. **Composite lift.** If $4/p = 1/x + 1/y + 1/z$ then $4/(mp) = 1/(mx) + 1/(my) + 1/(mz)$.
3. **Shallow-a divisor search.** For residual n, set $a \ge \lfloor n/4 \rfloor + 1$ and split the residual $p/q = (4a-n)/(na)$ into two unit fractions by enumerating divisors $d \mid q^2$ with $d \equiv -q \pmod p$ and $d \le q$ (ensuring $b \le c$).
4. **Unit tests.** Exhaustive check on 2..499; known $n = 5$; hard-ish prime 1009 (first prime missed by the mod-840 Mordell cover).

Results (numbers!)

Unit tests

`PASS`: range 2..499; $n = 5$; $n = 1009$.

Box verification

Quantity	Value
N **verified**	**1000000** (10^6)
Integers checked	999999
Failures	**0**
All verified	**True**

Quantity	Value
Identity hits	833333
Search / lift hits	166666
Wall total	**18.2732 s**

Sample solutions (a, b, c)

n	a	b	c
2	1	2	2
3	2	2	3
4	2	4	4
5	2	5	10
13	4	18	468
17	6	17	102
1009	253	85100	944524900
10001	3334	10001	33343334
500000	250000	500000	500000
1000000	500000	1000000	1000000

Calibration

Auto-selected N from ladder $\{10^4, 2 \cdot 10^4, 5 \cdot 10^4, 10^5, 2 \cdot 10^5, 5 \cdot 10^5, 10^6\}$ under a $\sim$ 2–3 minute wall budget (probe rate at 10^4). Achieved $N = 10^6$ in under 20 seconds.

Why Band-1 vs Band-4

- **Band-1 (this note):** shrink / certify an **explicit finite range**; exercise identities + search; ledger literature. Haystack / range collapse.
- **Band-4 / OPEN:** a proof (or counterexample) for **all** $n \geq 2$. Still open.

Honesty one-liner

Erdős–Straus verified for all $n \leq 10^6$ on box (0 failures); full conjecture remains OPEN.

Code

`/workspace/experiments/band1/erdos_straus/erdos_straus.py`